

Incident and Risk Review: Q3-17

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Incident overview

Quality Incident Q3-17 concerns Orion Control Modules from Helios Components production lot HC-447. Environmental verification found intermittent communication loss after high-humidity exposure.

The modules were intended for conveyor control cabinets assembled by Kestrel Logistics for Project Atlas. No units from the affected lot entered live customer operations, and no injury or customer-data event occurred.

Northstar classified the incident as Level 3 because it threatened a strategic program milestone and involved a sole-qualified component source.

Contents

- 1. Incident overview
- 2. Detection and timeline
- 3. Technical cause
- 4. Operational impact
- 5. Relationship and dependency analysis
- 6. Containment and corrective action
- 7. Residual risk and decisions
- 8. Lessons and follow-up

2. Detection and timeline

On 2 September, incoming inspection observed one intermittent failure. On 3 September, Quality Engineering reproduced the issue under humidity testing and quarantined lot HC-447. Kestrel Logistics stopped cabinet integration the same day.

Helios provided initial containment on 5 September and shipped replacement samples on 9 September. Replacement lot HC-452 completed incoming checks on 14 September.

Project Atlas revised integrated testing by twelve calendar days and moved its forecast pilot launch from 12 October to 24 October.

3. Technical cause

The investigation found inconsistent conformal-coating thickness near one connector family. The thin area allowed moisture-related signal instability under the environmental test profile.

Helios had adjusted a dispensing fixture during maintenance but did not classify the adjustment as a process change requiring Northstar notification. The inspection sample also did not target the affected connector area.

The root cause therefore combined an uncontrolled maintenance adjustment, incomplete change classification, and insufficient inspection coverage.

4. Operational impact

Northstar quarantined 186 modules. Kestrel Logistics paused six cabinet builds and resequenced other work. The Project Atlas team delayed integrated testing, while warehouse teams moved some training to classroom exercises.

The incident added expedited freight, inspection labor, and program-management effort. It did not affect the Q3 revenue table directly because Atlas was not yet a production revenue system.

The main business effect was schedule risk and continued dependence on a single qualified supplier.

5. Relationship and dependency analysis

Helios Components manufactures the Orion Control Module. Kestrel Logistics integrates that module into conveyor control cabinets. Project Atlas depends on the cabinets for its Bengaluru warehouse pilot.

Quality Incident Q3-17 affected Helios lot HC-447, which caused Kestrel to pause integration and delayed the Atlas testing milestone. Priya Nair owns the Atlas program; Marcus Lee is executive sponsor for both Atlas escalation and the Tier 1 vendor review.

This chain demonstrates why isolated text retrieval may find relevant passages yet miss the complete multi-hop relationship.

6. Containment and corrective action

Immediate containment included lot quarantine, expanded humidity testing, replacement samples, and a temporary shipment hold. Helios added coating-thickness checks at the affected connector area.

Permanent actions require fixture-change classification, revised maintenance approval, automated coating verification, operator retraining, and three conforming production lots. The full corrective-action package is due 18 October 2026.

Northstar Quality Engineering will close the action only after evidence review and successful validation; supplier declaration alone is insufficient.

7. Residual risk and decisions

Residual likelihood is medium until three conforming lots are verified. Program impact remains medium because replacement modules are available but alternate-source qualification is incomplete.

The steering committee may proceed with the 24 October pilot only after a five-day stability test, completion of safety checks, and confirmation that no HC-447 modules remain in pilot cabinets.

If validation fails, the fallback is reduced pilot scope or postponement. Quarantined modules must not be released through a schedule exception.

8. Lessons and follow-up

The review identified three lessons: maintenance adjustments can be meaningful process changes; relationship maps speed impact analysis; and single-source mitigation must be executable before an incident occurs.

Vendor Operations will verify Helios records weekly. Project Atlas will track alternate-source qualification as a steering metric. Quality Engineering will update inspection templates for moisture-sensitive control modules.

The incident will be reviewed again after three conforming lots or on 15 December 2026, whichever occurs first.